

Condition-Wise Sinkhorn Drifting for One-Shot Learned Channel Simulation

Rick Fritschek and Rafael F. Schaefer

Abstract—Learned communication systems may evaluate stochastic channel surrogates millions of times inside differentiable training loops. Conditional diffusion models provide strong sample quality, with inference cost set by reverse denoising steps. This paper proposes condition-wise Sinkhorn drifting, a one-shot surrogate that replaces unconditional or global transport with per-condition transport over the laws $p(y | x)$. We formulate a conditional Sinkhorn objective by integrating Sinkhorn divergences over fixed-input output laws and train a one-shot generator through detached particle updates. Experiments on additive white Gaussian noise (AWGN), Rayleigh fading, solid-state power amplifier (SSPA) nonlinearity, and a compact tapped-delay-line (TDL) channel compare the proposed method with conditional denoising diffusion probabilistic modeling (DDPM), denoising diffusion implicit modeling (DDIM), Wasserstein generative adversarial network (WGAN), and drifting ablations. Condition-wise Sinkhorn is the best-performing one-shot drifting-family surrogate under conditional diagnostics and symbolic-coding checks. Diffusion remains strongest on the hardest downstream symbol-error-rate (SER) curves, and global output-space sliced Wasserstein distance (SWD) alone is unreliable for model selection.

Index Terms—learned channel models, conditional generative modeling, diffusion models, drifting models, optimal transport, Sinkhorn divergence, Wasserstein gradient flows, inference latency

I. INTRODUCTION

Accurate channel models are a basic requirement for modern communication system design. They are particularly important in neural and end-to-end learned systems, where the channel appears repeatedly inside training loops, decoder adaptation procedures, and differentiable autoencoder pipelines. When a clean analytical channel model is unavailable, too restrictive, or poorly matched to measured data, generative channel surrogates become attractive because they can learn the full conditional distribution directly from samples.

This problem also involves more than distributional accuracy. In many learning pipelines, the channel model is evaluated millions of times. As a result, sample quality, conditioning accuracy, computational cost, and differentiability all matter simultaneously. The practical question is whether a generative model can match the channel distribution at a cost compatible with large-scale training and deployment workflows.

Recent work on diffusion-based channel modeling has shown that conditional diffusion models can generate high-quality channel samples across standard benchmarks such as additive white Gaussian noise (AWGN), Rayleigh fading, and nonlinear amplifier channels [1]. Follow-up work has further

studied robustness and speed-quality tradeoffs for diffusion-based channel generation [2], extended diffusion-based channel synthesis to high-dimensional user-specific channels [3], and proposed diffusion-driven digital-twin-style generation of statistical channel state information from sensing and location information [4]. More broadly, diffusion methods have also begun to influence adjacent wireless tasks such as high-dimensional channel estimation [5], low-complexity multiple-input multiple-output (MIMO) channel estimation [6], [7], joint channel estimation and data detection [8], and fast time-varying MIMO orthogonal frequency-division multiplexing (MIMO-OFDM) estimation [9]. This line of work is compelling because diffusion models are flexible, robust, and often more stable than earlier adversarial approaches. This improvement comes with a practical system cost. Inference requires an iterative reverse denoising procedure, and the latency scales with the number of diffusion steps. This tradeoff is manageable when the channel generator is called occasionally. It becomes more problematic when the generator is embedded in inner loops of iterative training or repeated Monte Carlo evaluations. This latency bottleneck has motivated several acceleration strategies for diffusion, including skipped or reduced-step sampling as in denoising diffusion implicit modeling (DDIM), as well as one-step or few-step alternatives such as consistency models and distillation-based approaches that compress multi-step samplers into substantially shorter generation paths [10]–[12]. This motivates the study of one-shot conditional generators for learned channel simulation. Among recently proposed generative paradigms, drifting models [13] are particularly interesting because they learn the transport from a simple latent source to the target data distribution directly in the network parameters. Intuitively, diffusion spends computation at test time by progressively transforming noise into data through a reverse-time chain. Drifting spends computation during training. Generated samples are repeatedly nudged toward the data manifold by a drift field, and the neural generator is updated so that future samples move closer to the desired distribution in a single forward pass. Once trained, sampling is therefore naturally one-shot. This perspective is also consistent with recent geometric analyses of generative dynamics that relate diffusion-like behavior to autonomous vector-field structure and marginalized noise geometry [14]. Recent work on one-step generation via Wasserstein gradient flows (W-Flow) [15] sharpens this viewpoint by replacing heuristic drifting fields with Sinkhorn-divergence optimal-transport velocities. This is directly relevant to channel simulation, because the quality and stability of the drift field is the main design question once inference is constrained to one generator evaluation.

The distinction between ordinary generation and channel

simulation is central. In ordinary generative modeling, samples are transported in the data space until the model distribution matches the target distribution. In channel simulation, the transmitted symbol is fixed side information. The relevant target is the family of conditional laws $\{p(\cdot | x)\}_{x \sim \mu}$. A generator can match the global output cloud while mixing conditional fibers, which produces an incorrect channel surrogate for communication-system training. This motivates a condition-wise transport formulation in which couplings preserve x and move only the output component y . The proposed condition-wise Sinkhorn drifting implements this idea as a one-shot neural channel generator.

The emphasis of this study is on generator-level distributional accuracy, computational tradeoffs, and downstream channel-implant checks. A full communication-system design study over all modulation and coding regimes is outside the present scope. The condition-wise Sinkhorn estimator also assumes simulator-style access to repeated outputs at the same transmitted symbol. Measured datasets with only one observation per condition require local conditional approximations, which we discuss as a limitation and future extension.

The main contributions are summarized as follows.

- We formulate condition-wise Sinkhorn drifting for learned channel simulation by integrating Sinkhorn divergences over fixed-input output fibers $p(\cdot | x)$.
- We connect this objective to a conditional Wasserstein-flow velocity, then state how the practical method differs from the population flow through finite-sample barycentric estimation and projected detached-target neural training.
- We implement conditional drifting in direct-output and residual parameterizations and compare target-kernel, conditioning-aware joint-kernel, joint Sinkhorn, and condition-wise Sinkhorn drift fields under a matched one-shot generator.
- We compare the proposed one-shot surrogate against conditional DDPM, conditional DDIM, and conditional WGAN reference models on AWGN, Rayleigh fading, SSPA, and compact TDL channel studies wherever the protocols overlap.
- We evaluate channel surrogates with direct SWD, anchor-conditioned moment metrics, downstream SER, bit error rate (BER), block error rate (BLER), and timing measurements, showing that communication usefulness is better reflected by conditional and downstream metrics than by global SWD alone.

Source code will be made available through the first author's GitHub repository upon publication¹.

II. PROBLEM FORMULATION AND DRIFTING BACKGROUND

A. What drifting changes conceptually

Because drifting is very recent and remains less familiar than diffusion, we briefly summarize its generative viewpoint.

Following [13], let f_θ map a latent variable $z \sim p_z$ to a sample $u = f_\theta(z)$, and let

$$q_\theta = (f_\theta)_\# p_z \quad (1)$$

denote the pushforward distribution induced by the generator. The training objective uses a *drifting field* that tells us how samples drawn from the current model distribution should move so that the model distribution approaches the target distribution p .

At a high level, the desired property is an equilibrium condition. If the model already matches the target distribution, then the drift should vanish. Using $V_{p,q}(u)$ to denote the field evaluated at a sample u under target distribution p and model distribution q , the fixed-point intuition can be summarized as

$$q = p \implies V_{p,q}(u) = 0, \quad (2)$$

which is the sense in which matched distributions form an equilibrium of the training dynamics [13].

This makes the training logic very different from diffusion. In diffusion, one learns a denoiser and then performs iterative refinement at inference time. In drifting, one instead constructs improved targets during training by moving current samples along the field $V_{p,q}$, and then regresses the generator onto those improved targets. As a result, the iterative correction happens during optimization.

The one-step view can be written as follows. Starting from a generated sample $u = f_\theta(z)$, define a drifted target

$$\tilde{u} = \text{stopgrad}(u + \eta V_{p,q_\theta}(u)), \quad (3)$$

and optimize the generator by

$$\mathcal{L}_{\text{drift}} = \mathbb{E}_{z \sim p_z} [\|f_\theta(z) - \tilde{u}\|_2^2]. \quad (4)$$

The stop-gradient is essential here because it makes the drifted sample behave like a fixed target for the current optimization step. Repeated training steps therefore evolve the pushforward q_θ toward the target distribution. This is also the cleanest way to see why drifting is naturally one-shot at inference. Once the generator has learned to emit samples near equilibrium, sampling only requires drawing z and evaluating $f_\theta(z)$, without a reverse-time chain.

The W-Flow formulation [15] connects this one-shot training principle to entropic optimal transport. Let $\text{OT}_\varepsilon(q, p)$ denote the entropic optimal-transport cost between a generated law q and target law p , computed with Sinkhorn iterations [16]. The debiased Sinkhorn divergence is

$$S_\varepsilon(q, p) = \text{OT}_\varepsilon(q, p) - \frac{1}{2} \text{OT}_\varepsilon(q, q) - \frac{1}{2} \text{OT}_\varepsilon(p, p), \quad (5)$$

which removes the entropic self-bias and interpolates between optimal-transport and kernel-like behavior as ε changes [17]. For an empirical generated particle u , the corresponding W-Flow velocity can be written in barycentric-projection form as

$$V_\varepsilon(u) = T_{q,p}^\varepsilon(u) - T_{q,q}^\varepsilon(u), \quad (6)$$

where $T_{q,p}^\varepsilon$ maps a generated sample to the barycenter of target samples under the entropic optimal-transport coupling, and $T_{q,q}^\varepsilon$ is the generated self-transport projection. This has the same attraction–repulsion structure as the practical drifting

¹https://github.com/Fritschek/conditional_drifting_models.

TABLE I
TWO TARGET PARAMETERIZATIONS FOR CONDITIONAL DRIFTING.

Mode	Output of g_θ	Target object	Short interpretation
Direct	$\hat{y}$	Learn $p(y x)$	Learns the channel law directly
Residual	$\hat{e}$, then $\hat{y} = x + \hat{e}$	Learn $p(e x)$ with $e = y - x$	Learns the channel perturbation

rule. The weights come from a globally mass-constrained coupling that replaces independently row-normalized kernels.

Fig. 1 gives a toy particle simulation in the spirit of the particle-based demonstrations used in [13]. For two fixed channel inputs, we initialize mismatched conditional residual particle clouds and then evolve them under the same attraction–repulsion drift rule used in our channel benchmark. The figure illustrates the drift dynamics in a controlled setting and serves as a toy analogue of the full learned generator. To show that the same idea can train an explicit parameterized generator, Fig. 2 uses a tiny conditional MLP trained with the drifting objective on the same toy target.

B. Conditional channel simulation

We now specialize the generic drifting view to learned conditional channel simulation. Let $x \in \mathbb{R}^n$ denote the transmitted channel input and $y \in \mathbb{R}^n$ the channel output. The objective is to learn a conditional generator that reproduces the channel law given x . In this setting, conditioning enters through the transmitted symbol, while stochasticity is carried by a latent variable $z \sim \mathcal{N}(0, I)$.

Two target parameterizations are useful in the experiments. The first is the direct channel-output formulation, in which the generator approximates $p(y | x)$ and predicts $\hat{y} = g_\theta(x, z)$. The second is the residual formulation, in which

$$e = y - x \quad (7)$$

is modeled instead, so that the generator approximates $p(e | x)$ and the channel output is reconstructed as

$$\hat{y} = x + \hat{e}. \quad (8)$$

The direct formulation learns the channel law itself. The residual formulation injects the inductive bias that the channel output can be understood as the transmitted symbol plus a structured perturbation. In the experiments below, the direct formulation aligns naturally with direct-output diffusion and WGAN baselines. The residual formulation is often easier to optimize, although it can fall short on nonlinear channels.

C. Conditional drifting model

We use a practical conditional adaptation inspired by [13]. To avoid confusion with the communication input symbol x , we use u above for the generic sample variable in the abstract formulation and reserve x below for the transmitted symbol. In our implementation, drifting can be applied either in direct-output space or in residual space. In direct mode, the

generator predicts $\hat{y} = g_\theta(x, z)$. In residual mode, it predicts $\hat{e} = g_\theta(x, z)$ with reconstruction $\hat{y} = x + \hat{e}$, where

$$z \sim \mathcal{N}(0, I). \quad (9)$$

In both cases, g_θ is implemented as a two-hidden-layer multi-layer perceptron with hidden width 128, latent dimension 16, and SiLU activations.

$$[x, z] \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear} \rightarrow \text{SiLU} \rightarrow \text{Linear}. \quad (10)$$

The training target is formed from an attraction–repulsion drift field in the chosen target space. This is a concrete channel-specific instantiation of the generic field $V_{p,q}$. Given generated samples $G = \{g_i\}$ and positive samples $P = \{p_j\}$, we compute

$$V(g_i) = \left(\sum_j w_{ij}^{(P)} p_j - g_i \right) - \lambda \left(\sum_k w_{ik}^{(G)} g_k - g_i \right), \quad (11)$$

where $w^{(P)}$ and $w^{(G)}$ are row-normalized RBF kernel weights. The first term attracts each generated sample toward a local barycenter of true samples in the chosen target space, and the second term repels it from nearby generated samples to reduce collapse. This yields a detached target

$$\tilde{g}_i = \text{stopgrad}(g_i + \eta V(g_i)), \quad (12)$$

and the generator is optimized by

$$\mathcal{L}_{\text{drift}} = \frac{1}{|G|} \sum_i \|g_i - \tilde{g}_i\|_2^2. \quad (13)$$

In other words, the generic update $u \mapsto u + \eta V_{p,q}(u)$ from the drifting formulation is realized here with a kernel-based drift field estimated directly from minibatches of true and generated samples in the selected target space. At a high level, this empirical drift field can be interpreted as a kernel particle interaction rule. The attraction term pulls generated samples toward the target sample cloud, while the repulsion term prevents generated particles from collapsing onto a few modes. In this sense, the method is related in spirit to kernel-based distribution shaping and MMD-like matching. Here it is used as a practical training rule, without deriving a formal discrepancy minimization objective.

In the simulations, the drift scale is $\eta = 1.0$, the repulsive weight is $\lambda = 1.0$, the minimum kernel bandwidth is 0.2, and the maximum drift norm is 2.0. These settings were chosen because they produced stable training behavior and materially improved the early AWGN drifting baselines.

a) *Conditioning-aware kernel variant.*: For the enhanced direct-output drifting model, we replace the target-only kernel by a kernel on joint condition-target pairs. Instead of computing sample interactions from $k(y_i, y_j)$, we compute them from $k((x_i, y_i), (x_j, y_j))$, so that attraction and repulsion depend on similarity in both the channel input and the output space. In the direct-output setting, the target kernel can also incorporate residual similarity, yielding a drift field that is more closely aligned with the conditional channel structure. The generator architecture is unchanged.

This kernel rule is the conditioning-aware baseline used for the direct-output drifting variants. It is computationally simple. It also leaves the most important design choice—the minibatch coupling used to build V —as a channel-specific heuristic. We therefore also study Sinkhorn-based drift fields that replace the row-normalized kernel interaction with the velocity in (6), while preserving the same detached-target training loop and the same one-shot inference path.

D. Condition-wise Sinkhorn drifting

The W-Flow objective in [15] is unconditional. Both p and q_θ are distributions on the generated sample space. For channel simulation, the generated and true joint laws share the same transmitted-symbol marginal. Writing μ for the input distribution,

$$p(dx, dy) = \mu(dx)p_x(dy), \quad q_\theta(dx, dy) = \mu(dx)q_{\theta,x}(dy). \quad (14)$$

The condition x is fixed side information. Transporting in the full joint space (x, y) therefore solves a different problem from matching the channel law $p(y | x)$, because it allows the geometry of the fixed condition variable to influence the coupling while the update can only move the output component.

We instead define the conditional Sinkhorn functional

$$\mathcal{S}_\varepsilon^{\text{cond}}(q_\theta, p) = \int S_\varepsilon(q_{\theta,x}, p_x) \mu(dx), \quad (15)$$

where the Sinkhorn divergence is computed separately for each fixed transmitted symbol, either on y in direct-output mode or on $e = y - x$ in residual mode. In measure-theoretic terms, the conditional law $p_x = p(\cdot | x)$ is the output fiber over condition x . Throughout the paper we refer to the resulting estimator as condition-wise Sinkhorn drifting. Equivalently, admissible couplings must preserve the condition.

$$\pi(dx, dy, d\tilde{y}) = \mu(dx)\pi_x(dy, d\tilde{y}), \quad \pi_x \in \Pi(q_{\theta,x}, p_x). \quad (16)$$

The Wasserstein gradient flow of (15) has zero x -velocity and evolves only inside each fixed-input conditional law.

$$\partial_t q_t(x, y) + \nabla_y \cdot (q_t(x, y)v_t(x, y)) = 0. \quad (17)$$

Applying the W-Flow velocity condition-wise gives

$$v_t(x, y) = T_{q_t,x,p_x}^\varepsilon(y) - T_{q_t,x,q_t,x}^\varepsilon(y). \quad (18)$$

This interpretation is deliberately limited to the population conditional optimal-transport flow. For μ -almost every x , assume that p_x and $q_{t,x}$ are Borel probability measures on $\mathbb{R}^n$ with finite second moments and that the entropic OT potentials are differentiable along the considered distributional curve. Under the standard Sinkhorn-divergence assumptions used in [17], $S_\varepsilon(q_{t,x}, p_x) \geq 0$ and equality holds exactly when $q_{t,x} = p_x$. The exact Wasserstein gradient flow then satisfies

$$\frac{d}{dt} \mathcal{S}_\varepsilon^{\text{cond}}(q_t, p) = - \int \|v_t(x, \cdot)\|_{L^2(q_{t,x})}^2 \mu(dx) \leq 0. \quad (19)$$

Thus, if the condition-wise Sinkhorn velocity has equality as its only stationary point for each fixed condition, then every stationary point of the conditional flow satisfies

$$q_{t,x} = p_x \quad \text{for } \mu\text{-a.e. } x. \quad (20)$$

TABLE II
CONDITION-WISE SINKHORN DRIFTING UPDATE.

Input: generator g_θ , channel sampler $p(\cdot | x)$, anchors $\{x_i\}_{i=1}^B$, generated count K_g , positive count K_p , reference count K_r , drift scale η .

1. Draw $\hat{y}_{i,k} = g_\theta(x_i, z_{i,k})$, $k = 1, \dots, K_g$.
2. Draw positive samples $y_{i,j} \sim p(\cdot | x_i)$, $j = 1, \dots, K_p$.
3. Draw generated reference samples $\hat{y}'_{i,\ell} = g_\theta(x_i, z'_{i,\ell})$, $\ell = 1, \dots, K_r$.
4. For each anchor i , solve two output-space Sinkhorn problems, $\hat{Q}_i \leftrightarrow P_i$ and $\hat{Q}_i \leftrightarrow \hat{Q}'_i$.
5. Form barycentric velocity $v_{i,k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i,k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i,k})$.
6. Regress $g_\theta(x_i, z_{i,k})$ toward $\text{stopgrad}(\hat{y}_{i,k} + \eta v_{i,k})$.

This is the conditional analogue needed for channel simulation. Its equilibrium is equality of the complete family of conditional channel laws. Matching a joint sample cloud under an arbitrary joint cost leaves this condition-wise equality unenforced.

The implemented algorithm adds two approximations to this population object. First, it estimates the barycentric velocity from finite minibatch Sinkhorn couplings. Second, it projects the explicit particle update back into the neural generator class through detached-target regression. The resulting update differs from exact gradient descent on $\mathcal{S}_\varepsilon^{\text{cond}}$ with respect to θ . If the generator family cannot realize all conditional laws p_x , the practical fixed point is the best representable conditional surrogate reached by this projected training dynamics. For anchors $x_i \sim \mu$, the estimator draws multiple generated samples $\hat{y}_{i,k} = g_\theta(x_i, z_{i,k})$, multiple positive channel samples $y_{i,j} \sim p(\cdot | x_i)$, and an independent generated reference batch $\hat{y}'_{i,\ell} = g_\theta(x_i, z'_{i,\ell})$. Sinkhorn barycentric projections are then computed separately for each anchor:

$$v_{i,k} = T_{\hat{Q}_i, P_i}^\varepsilon(\hat{y}_{i,k}) - T_{\hat{Q}_i, \hat{Q}'_i}^\varepsilon(\hat{y}_{i,k}), \quad (21)$$

followed by the detached regression target

$$\tilde{y}_{i,k} = \text{stopgrad}(\hat{y}_{i,k} + \eta v_{i,k}). \quad (22)$$

Table II summarizes the resulting training update.

This estimator is exact for simulator channels where repeated draws at the same x_i are available. For measured channels with only one observation per condition, the same idea requires a local conditional approximation, for example by grouping nearby x -values or using condition-space kernel weights before solving the output-space Sinkhorn problem.

The major operational advantage is straightforward. Once training is complete, sampling requires only a single evaluation of $g_\theta(x, z)$. In residual mode, this produces a channel residual sample $\hat{e}$ followed by reconstruction $\hat{y} = x + \hat{e}$. In direct mode, it produces $\hat{y}$ directly. Sampling avoids reverse chains, ODE solvers, and per-sample iterative schedules at inference.

III. EXPERIMENTAL SETUP

A. Channel models

We use four stochastic channel families. All channels are implemented as real-valued maps $x \mapsto y$, but the coordinate interpretation differs by channel. AWGN and Rayleigh use real-valued vectors directly and have dimension $n = 7$ in

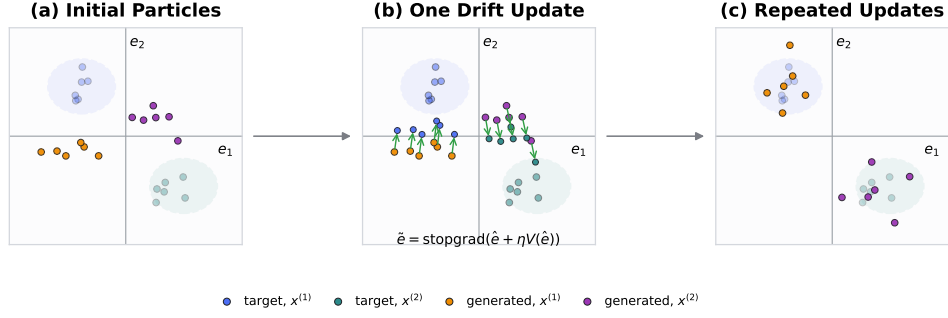


Fig. 1. Toy conditional particle simulation in residual space. Two different channel inputs induce two different target residual distributions. Starting from mismatched generated particle clouds, the middle panel applies one attraction–repulsion drift update of the form $\hat{e} = \text{stopgrad}(\hat{e} + \eta V(\hat{e}))$, and the last panel shows the particle clouds after repeated updates. The figure is an actual toy simulation of the drift field.

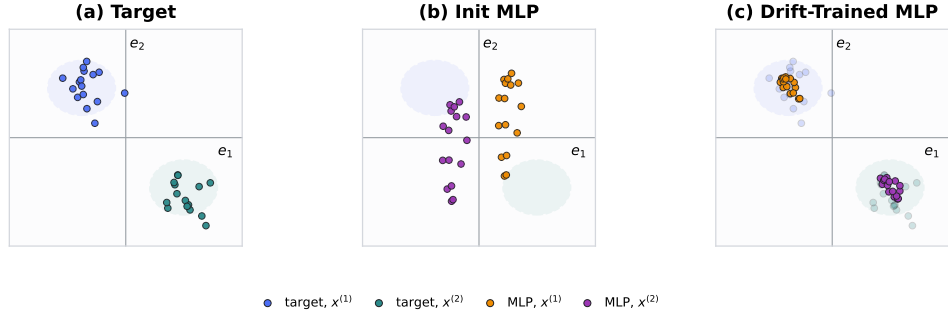


Fig. 2. Toy conditional MLP trained with the drifting objective. The left panel shows target conditional residual samples. The middle panel shows samples from a randomly initialized tiny conditional MLP $g_\theta(x, z)$. The right panel shows the same MLP after training with the drift objective, where the generated samples move close to the correct condition-dependent residual clouds.

the generator-level benchmark. SSPA and TDL use paired in-phase/quadrature coordinates with $n = 8$, corresponding to four complex symbols. Noise variables are sampled independently across real coordinates unless the deterministic channel transformation or tapped-delay convolution introduces coupling. The original generator-level comparison uses AWGN, Rayleigh, and SSPA because these channels have matched DDPM, DDIM, WGAN, and drifting reference runs. The downstream channel-implant study additionally includes TDL as a wireless channel with memory. For this channel, the DDPM, DDIM, and WGAN direct-output SWD rows are evaluated from the trained baseline checkpoints used in the SER-curve study.

AWGN:

$$y = x + n, \quad n \sim \mathcal{N}(0, \sigma^2 I). \quad (23)$$

This is the standard additive white Gaussian noise reference channel and serves as the simplest near-identity baseline.

Rayleigh: the transmitted symbol is multiplied elementwise by a random fading amplitude and then corrupted by additive Gaussian noise:

$$y = h \odot x + n, \quad (24)$$

where $\odot$ denotes elementwise multiplication,

$$h_k = \frac{1}{\sqrt{2}} \sqrt{u_k^2 + v_k^2}, \quad u_k, v_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1), \quad (25)$$

and $n \sim \mathcal{N}(0, \sigma^2 I)$.

Each real component is therefore scaled by an independent Rayleigh-distributed magnitude before noise is added. This multiplicative channel is correspondingly less aligned with a pure residual perturbation view. As implemented here, Rayleigh is a real-valued vector channel. The paired-I/Q complex representation is used for SSPA and TDL.

SSPA: the input is passed through a smooth solid-state power amplifier nonlinearity before additive noise is applied. In this case, the real-valued representation is interpreted as paired I/Q components. Let

$$r = \sqrt{x_I^2 + x_Q^2} \quad (26)$$

denote the input amplitude. The nonlinear gain factor used in the benchmark is

$$a(r) = \frac{G}{\left(1 + \left(\frac{Gr}{A_0}\right)^{2p}\right)^{1/(2p)}}, \quad (27)$$

with default parameters $p = 3$, $A_0 = 1.5$, and $G = 5.0$. Thus the output amplitude is $a(r)r$, matching the Rapp-style AM/AM law $Gr/(1 + (Gr/A_0)^{2p})^{1/(2p)}$. The noiseless amplifier output is $\tilde{x} = a(r)x$, and the observed channel output is

$$y = \tilde{x} + n, \quad n \sim \mathcal{N}\left(0, \frac{\sigma^2}{2} I\right). \quad (28)$$

This channel introduces amplitude-dependent nonlinear distortion while preserving the input direction in the I/Q plane.

TDL: The TDL channel is a compact short-block approximation of the 3GPP TR 38.901 TDL-D profile [18]. The real vector is interpreted as a complex codeword $s_m = x_{2m} + jx_{2m+1}$, $m = 0, \dots, N_c - 1$, with $N_c = 4$ complex channel uses in the symbolic coding runs. For each codeword, the channel samples a finite set of complex taps and applies circular convolution,

$$\tilde{s}_m = \sum_{\ell=0}^{L_h-1} h_\ell s_{(m-d_\ell) \bmod N_c}. \quad (29)$$

The observed output is

$$y_m = \tilde{s}_m + n_m, \quad n_m \sim \mathcal{CN}(0, \sigma^2). \quad (30)$$

The implemented profile uses the TDL-D line-of-sight structure with first-tap K -factor 13.3 dB. Because the codeword is short, the normalized path delays from the standard profile are rounded to a small set of effective circular shifts and path powers are renormalized after aggregation. This keeps the channel compatible with the short symbolic autoencoder while preserving the random multipath mixture structure. We use this TDL configuration as a compact channel-with-memory stress test. It is intentionally smaller than a full OFDM or long-block TDL evaluation.

B. Training protocol

The experiments are organized into two complementary protocols. First, the generator-level protocol compares the proposed drifting family against the established conditional DDPM, DDIM, and WGAN references on AWGN, Rayleigh, and SSPA. Second, the drift-field ablation holds the drifting network fixed and compares target-space kernel drifting, conditioning-aware joint-kernel drifting, joint Sinkhorn drifting, and condition-wise Sinkhorn drifting. This second protocol is reported over 100 seeds for the main drift-field runs and is evaluated with direct SWD, anchor-conditioned metrics, and downstream symbolic SER/BER. It is the controlled comparison in this paper because the generator architecture, optimizer family, and one-shot inference path are held fixed across drift fields. The DDPM, DDIM, and WGAN rows provide best-effort reference baselines using stable model-family settings. A matched-capacity ranking against drifting would require an additional equal-parameter and equal-budget study. For SSPA, we additionally report a 30-seed update-budget-controlled condition-wise Sinkhorn run because the corrected same-condition transport field saturates after far fewer optimizer updates than the long diffusion-style SSPA preset. The TDL channel is included in the drift-field study and in the downstream SER curves with diffusion and WGAN references. For the direct-output SWD table, TDL W-Flow rows come from the drift-field ablation, and TDL DDPM/DDIM/WGAN rows are obtained by checkpoint-only direct-SWD evaluation of the corresponding trained baseline implants.

a) Channel and data settings.: Table III collects the channel dimensions, message alphabets, training points, and generator-evaluation budgets. The noise standard deviations induced by these $(E_b/N_0, R)$ pairs are 0.5260, 0.2350, 0.3251, and 0.3162 for AWGN, Rayleigh, SSPA, and TDL. The SSPA

condition-wise Sinkhorn row uses the compact update-budget screen because the corrected same-condition field had already converged at this budget. Continuing to the full 10^7 -sample SSPA preset over-optimized this sharper field. Downstream symbolic autoencoders are trained through either the analytic channel or a learned channel implant and are always evaluated on the analytic channel.

b) Model settings.: Drifting uses a conditional MLP with latent dimension 16, hidden width 128, two hidden layers, SiLU activations, and Adam with learning rate 10^{-3} . The base drifting objective uses drift scale 1.0, median-heuristic bandwidth selection, minimum bandwidth 0.2, maximum drift norm 2.0, and repulsive weight 1.0. We train both parameterizations, direct drifting with target y and residual drifting with target $e = y - x$. The four drift-field variants use the same direct-output generator architecture and detached-target update, but replace or modify the training drift field. The target-kernel row uses only target-space similarities, the joint-kernel row uses the conditioning-aware kernel, the joint Sinkhorn row uses a Sinkhorn/W-Flow drift in joint condition-output features, and the condition-wise row uses the per-anchor conditional Sinkhorn field in (18). Unless otherwise stated, Sinkhorn variants use 10 Sinkhorn iterations, automatic entropic scale selection with minimum $\varepsilon = 10^{-3}$, four generated samples, four positive channel samples, and four generated reference samples per condition anchor.

For AWGN, Rayleigh, and SSPA, diffusion models $p(y | x)$ directly with v -prediction, cosine-ZF variance schedule, 100 diffusion steps, and EMA decay 0.9. Hidden widths are 110 (AWGN), 128 (Rayleigh), and 110 (SSPA). AWGN and Rayleigh use a two-stage learning-rate schedule (10 epochs at 10^{-3} , then 20 epochs at 10^{-4}), while SSPA uses a constant 10^{-4} for 160 epochs. We report DDPM and DDIM with 100, 50, 20, and 10 denoising steps.

WGAN uses a conditional Wasserstein objective with RM-Sprop at 10^{-4} for generator and critic, five critic updates per generator update, and weight clipping at 0.01. Hidden widths are 128 (AWGN) and 256 (Rayleigh/SSPA). WGAN is reported in direct output space wherever it is compared against direct-output channel generators.

c) Experimental fairness.: The benchmark is intended as a practical comparative study, with stable model-family implementations prioritized over strict matched-capacity control. Architectures and training schedules were chosen as stable implementations for each model family. Identical parameter counts and compute budgets across drifting, diffusion, and WGAN are outside the benchmark design. The Sinkhorn rows are therefore reported as a drift-field ablation under a matched drifting architecture and training budget. They are compared to diffusion and WGAN wherever the observable metric and channel protocol overlap. The downstream W-Flow table focuses on the learned drifting/Sinkhorn surrogates and the analytic-channel reference floor. The SER curves add WGAN and DDIM-100 when all corresponding channel implants are available.

As an additional diagnostic, Table IV reports inference-network parameter counts used in this benchmark.

TABLE III

SIMULATION SETUP USED FOR THE JOURNAL EXPERIMENTS. THE LISTED E_b/N_0 VALUES ARE THE NOMINAL CHANNEL-MODEL AND AUTOENCODER TRAINING POINTS. GENERATOR-LEVEL AWGN/RAYLEIGH/SSPA REFERENCE ROWS USE TEN SEEDS. W-FLOW ROWS USE 100 SEEDS EXCEPT FOR THE 30-SEED SSPA UPDATE-BUDGET SCREEN. ALL SWD EVALUATIONS USE 128 RANDOM PROJECTIONS.

Channel	n	M_{msg}	R	E_b/N_0	Train samples / batch / epochs	Eval. samples	Main use
AWGN	7	16	4/7	5 dB	$10^7/5000/30$	10^6	Reference + W-Flow
Rayleigh	7	16	4/7	12 dB	$10^7/5000/30$	10^6	Reference + W-Flow
SSPA reference	8	64	6/8	8 dB	$10^7/4096/160$	10^6	DDPM/DDIM/WGAN + direct drifting
SSPA W-Flow screen	8	64	6/8	8 dB	$1.2 \times 10^5/4096/160$	10^6	Reported condition-wise Sinkhorn
TDL	8	16	4/8	10 dB	$1.2 \times 10^5/512/60$	10^5	W-Flow + downstream curves

TABLE IV

MODEL PARAMETER COUNTS IN THE BENCHMARK SETUP. COUNTS REFER TO INFERENCE-TIME NETWORKS. DDPM AND DDIM SHARE THE SAME DENOISER WEIGHTS. WGAN ROWS REPORT GENERATOR-SIDE PARAMETERS.

Model	AWGN	Rayleigh / SSPA
Drifting (direct / residual)	20,487	20,487 / 20,744
Diffusion denoiser (DDPM / DDIM)	59,847	74,247 / 60,178
WGAN generator	19,335	71,431 / 72,200

d) *Seeding and reporting.*: The diffusion/WGAN reference benchmark results are reported as mean $\pm$ standard deviation over ten seeds. The Sinkhorn drift-field ablation is reported over 100 seeds unless stated otherwise. The SSPA update-budget check uses 30 seeds. We use standard deviation for the ten-seed reference rows to describe run-to-run variability. W-Flow tables use mean $\pm$ standard error because the main purpose of those rows is to compare seed-averaged drift-field variants. Each run uses deterministic initialization to reduce run-to-run variance unrelated to model stochasticity.

C. Evaluation metric

The diffusion/WGAN/drifting reference comparison emphasizes generator-level distributional accuracy and timing, while the Sinkhorn drift-field ablation adds anchor-conditioned diagnostics and downstream symbolic SER/BER. The primary metric is sliced Wasserstein distance (SWD), computed in the target space used by each benchmark path. For direct-output models, we report $\text{SWD}(\{\hat{y}_i\}, \{y_i\})$. For residual models, we report $\text{SWD}(\{\hat{e}_i\}, \{e_i\})$. Given true samples $P = \{v_i\}_{i=1}^N$ and generated samples $Q = \{\hat{v}_i\}_{i=1}^N$, SWD is estimated by drawing random unit directions θ_m , projecting both clouds onto those directions, computing the one-dimensional Wasserstein distance for each projection, and averaging over projections:

$$\text{SWD}(P, Q) = \frac{1}{M} \sum_{m=1}^M W_1(\{\langle \theta_m, v_i \rangle\}_{i=1}^N, \{\langle \theta_m, \hat{v}_i \rangle\}_{i=1}^N).$$

Thus SWD compares the generated and true sample clouds through many one-dimensional views and captures geometric mismatch more faithfully than a purely bin-based marginal comparison. In our experiments, SWD is estimated using 128 random one-dimensional projections on AWGN, Rayleigh, SSPA, and TDL. The projection directions are sampled from an isotropic Gaussian distribution, normalized to unit length, and generated from the deterministic metric seed associated with the run. SWD is used primarily for comparability with

prior diffusion-channel benchmarks [1], [2]. Compared with histogram-based L_1 distance, it also better captures sample geometry in $\mathbb{R}^n$. SWD provides a geometry-aware comparison of global output sample clouds. It leaves decision-relevant behavior such as error rates unmeasured. Those depend on the joint structure of (x, y) . For communication use, two generators with similar global sample-cloud discrepancy can induce different conditional means, conditional covariance errors, and downstream symbol-error rates. The Sinkhorn drift-field study therefore also includes anchor-conditioned moment metrics and symbolic channel-implant SER/BER checks when evaluating the proposed drifting variants. We interpret the metrics in the following order. Downstream SER, BER, and block error rate (BLER) under analytic-channel evaluation are the closest checks of communication usefulness. Anchor-conditioned metrics are the next diagnostic because they probe repeated outputs at fixed transmitted symbols. Global direct-output SWD is mainly a generator-level comparability metric. Residual-space SWD is kept as a parameterization diagnostic for residual-mode models.

IV. BENCHMARK RESULTS

A. Unified SWD benchmark across channels

Table V reports a unified direct-output SWD comparison across AWGN, Rayleigh, SSPA, and TDL. The reference rows summarize the diffusion and WGAN baselines. The drifting-family rows contain direct drifting and the proposed drift-field variants under a matched one-shot architecture. The controlled conclusion from this table is the ranking inside the drifting-family block, while the diffusion and WGAN rows provide reference context from their own stable training protocols. DDIM-100 gives the lowest direct-space SWD on AWGN, Rayleigh, and SSPA among the matched reference rows. Among the W-Flow variants, condition-wise Sinkhorn is strongest under direct SWD on AWGN, Rayleigh, and TDL, and is numerically close to the best SSPA W-Flow rows under the compact update-budget run. Within the full drifting-family block, condition-wise Sinkhorn is lowest on AWGN, Rayleigh, and TDL, while direct drifting remains lowest on SSPA under this global metric. This already indicates that drift-field ranking depends on channel geometry.

Timing follows the expected pattern. Drifting and WGAN operate at roughly $0.1 \mu\text{s}$ per sample on the local CUDA timing setup, while DDPM and DDIM require microsecond-to-tens-of-microseconds sampling times depending on the sampler and the number of denoising steps. Thus, even when diffusion

TABLE V

DIRECT-OUTPUT SWD COMPARISON FOR DIFFUSION/WGAN REFERENCES AND DRIFTING VARIANTS. LOWER IS BETTER. REFERENCE ROWS REPORT MEAN $\pm$ STANDARD DEVIATION FROM THE DIFFUSION/WGAN BENCHMARK OR FROM CHECKPOINT-ONLY DIRECT-SWD EVALUATION. THE DRIFTING-FAMILY ROWS CONTAIN DIRECT DRIFTING FROM THE SAME BENCHMARK AND THE W-FLOW DRIFT-FIELD ABLATION. W-FLOW ROWS REPORT THE AVAILABLE-SEED MEAN $\pm$ STANDARD ERROR. BOLD MARKS THE BEST DRIFTING-FAMILY ROW PER CHANNEL UNDER THE REPORTED MEAN. DASHES MARK UNAVAILABLE MATCHED GENERATOR-LEVEL SWD VALUES.

Method	AWGN	Rayleigh	SSPA	TDL
<i>Diffusion/WGAN reference baselines</i>				
WGAN	0.0198 $\pm$ 0.0050	0.0171 $\pm$ 0.0050	0.0287 $\pm$ 0.0156	0.0289 $\pm$ 0.0063
DDPM	0.0070 $\pm$ 0.0004	0.0079 $\pm$ 0.0008	0.0032 $\pm$ 0.0005	0.0297 $\pm$ 0.0011
DDIM-100	0.0042 $\pm$ 0.0006	0.0044 $\pm$ 0.0007	0.0024 $\pm$ 0.0004	0.0248 $\pm$ 0.0009
<i>Drifting family</i>				
Direct drifting	0.0100 $\pm$ 0.0007	0.0085 $\pm$ 0.0012	0.0060 $\pm$ 0.0008	–
Kernel target	0.0106 $\pm$ 0.0001	0.0099 $\pm$ 0.0001	0.0070 $\pm$ 0.0001	0.0413 $\pm$ 0.0003
Kernel joint	0.0290 $\pm$ 0.0004	0.0492 $\pm$ 0.0002	0.0074 $\pm$ 0.0001	0.0289 $\pm$ 0.0003
Joint Sinkhorn	0.0148 $\pm$ 0.0002	0.0152 $\pm$ 0.0002	0.0117 $\pm$ 0.0002	0.0966 $\pm$ 0.0005
Condition-wise Sinkhorn	0.0058 $\pm$ 0.0001	0.0073 $\pm$ 0.0001	0.0071 $\pm$ 0.0002	0.0118 $\pm$ 0.0002

achieves the strongest SWD, one-shot generators retain a substantial practical latency advantage. The W-Flow variants retain the same one-shot inference path as direct drifting. Their cost differences are training-time drift-field costs. Table VI therefore reports W-Flow ablation rows under the same 2% local CUDA timing protocol. The timing run uses PyTorch on a local NVIDIA RTX 5060 Ti GPU in standard float32 execution, without CUDA graph capture, `torch.compile`, or mixed precision. Training time is measured by running the actual training loop on 2% of the configured dataset size, rounded to full batches, and extrapolating by the ratio of timed optimizer steps to full optimizer steps. This includes synthetic channel sampling, forward and backward passes, optimizer updates, and the Sinkhorn computations used by W-Flow rows. It excludes checkpoint writing and downstream symbolic autoencoder training. Inference timing uses batch size 512, 40 batches per repeat, two warmup repeats, seven timed repeats, and CUDA synchronization around each repeat. It includes drawing the conditioning inputs and evaluating the generator. For diffusion rows, inference includes the complete DDPM or DDIM denoising trajectory and the same conditioning overhead. The total column adds projected training time to the time needed to generate the evaluation-sample budget listed in Table III. On AWGN, Rayleigh, and SSPA, the condition-wise Sinkhorn row is faster to train than the global kernel rows because it solves many small same-condition transport problems instead of forming a global minibatch interaction matrix.

B. Condition-wise drift-field interpretation

The W-Flow-inspired ablation isolates the drift-field design problem. The target-space kernel baseline, conditioning-aware joint kernel, naive joint Sinkhorn drift, and proposed condition-wise Sinkhorn drift all use the same one-shot generator architecture and differ only in how training particles are moved. The key comparison is therefore geometric. Joint Sinkhorn builds a transport field from (x, y) -features, while condition-wise Sinkhorn solves separate output-space transport problems over fixed x -fibers. This distinction is visible in the mixed metric behavior. Condition-wise Sinkhorn is the

strongest W-Flow row under direct SWD on AWGN, Rayleigh, and TDL, while SSPA needs the compact update-budget run because the corrected same-condition field converges earlier than the long diffusion-style preset. Fig. 3 shows the corresponding fixed-condition diagnostic for SSPA. For three transmitted-symbol anchors, the joint Sinkhorn field misses both the conditional mean and the local spread of the analytic output fiber. Condition-wise Sinkhorn keeps the generated cloud close to the analytic same- x cloud. The downstream checks below are the decisive test for this channel-simulator setting. They show why global SWD should be treated as a comparability diagnostic and supplemented by conditional or downstream metrics for model selection.

C. Conditional metrics and symbolic coding

Table VII reports the corresponding anchor-conditioned diagnostics. The anchor metrics draw repeated outputs at fixed transmitted symbols and therefore probe the conditional law more directly than global SWD. Direct SWD is kept in Table V, and downstream BER/SER is kept in Table VIII. This avoids reporting the same quantities in multiple tables. For AWGN, Rayleigh, SSPA, and TDL, condition-wise Sinkhorn is the best-performing drifting-family surrogate under the downstream BER/SER checks reported here. The full curve comparison below shows that diffusion remains the strongest learned reference on Rayleigh, SSPA, and TDL. On SSPA, the anchor-conditioned metrics also explain why the compact update-budget run is preferable. The same-condition Sinkhorn field matches fixed-input output clouds well, while the full-budget corrected run over-optimizes the sharp field. This indicates that condition-wise transport is useful and that its optimizer budget must respect the geometry of saturated nonlinear channels.

The downstream symbolic results in Table VIII use the learned channel surrogate for autoencoder training and the analytic channel for final evaluation. The analytic row is therefore an analytic reference floor. We treat SER as the primary symbolic metric because the autoencoder message alphabet is finite. BER is retained in the table as an additional bit-level diagnostic. Within the W-Flow ablation table,

TABLE VI
PROJECTED TRAINING AND INFERENCE TIMING ON AN NVIDIA RTX 5060 Ti GPU. TRAINING HOURS ARE EXTRAPOLATED FROM A 2% TIMING RUN. INFERENCE IS REPORTED AS TIME PER SAMPLE.

Method	AWGN			Rayleigh			SSPA			TDL		
	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]	Train [h]	Inf.	Total [h]
<i>One-shot generators</i>												
Drifting (res.)	1.309	0.131 us	1.310	1.300	0.123 us	1.300	5.656	0.121 us	5.656	0.013	0.124 us	0.013
Drifting (dir.)	1.299	0.114 us	1.300	1.299	0.117 us	1.300	5.656	0.118 us	5.656	0.007	0.116 us	0.007
Joint-kernel drift	1.863	0.116 us	1.864	1.864	0.116 us	1.864	8.112	0.119 us	8.112	0.008	0.115 us	0.008
Joint Sinkhorn	0.917	0.116 us	0.918	0.919	0.115 us	0.920	3.956	0.117 us	3.956	0.010	0.116 us	0.010
Condition-wise Sinkhorn	0.048	0.115 us	0.048	0.048	0.115 us	0.049	0.296	0.117 us	0.297	0.009	0.115 us	0.009
WGAN	0.064	0.108 us	0.064	0.085	0.108 us	0.086	0.478	0.107 us	0.478	0.022	0.108 us	0.022
<i>Diffusion samplers</i>												
DDPM	0.020	0.046 ms	0.147	0.020	0.045 ms	0.145	0.124	0.047 ms	0.253	0.005	0.045 ms	0.007
DDIM-100	0.020	0.039 ms	0.128	0.020	0.039 ms	0.128	0.124	0.039 ms	0.231	0.005	0.039 ms	0.006
DDIM-50	0.020	0.019 ms	0.073	0.020	0.019 ms	0.073	0.124	0.019 ms	0.177	0.005	0.019 ms	0.006
DDIM-20	0.020	0.008 ms	0.042	0.020	0.008 ms	0.042	0.124	0.008 ms	0.145	0.005	0.008 ms	0.006
DDIM-10	0.020	0.004 ms	0.031	0.020	0.004 ms	0.031	0.124	0.004 ms	0.135	0.005	0.004 ms	0.005

All rows use the same 2% extrapolation protocol. In the one-shot block, bold total times mark the lowest projected total for each channel. The W-Flow rows use the same one-shot generator architecture as direct drifting. Their training-time differences come only from the drift-field computation.

TABLE VII
ANCHOR-CONDITIONED DIAGNOSTICS FOR W-FLOW VARIANTS. ANCHOR SWD AND GAUSSIAN WASSERSTEIN-2 (GW2) ARE COMPUTED FROM REPEATED SAMPLES AT FIXED CHANNEL INPUTS. DIRECT SWD AND BER/SER ARE REPORTED SEPARATELY IN TABLES V AND VIII. VALUES ARE SEED MEANS. LOWER IS BETTER.

Channel	Variant	Anchor SWD	Anchor GW2
AWGN	Kernel target	0.3711	1.2565
AWGN	Kernel joint	0.1425	0.5936
AWGN	Joint Sinkhorn	0.1199	0.4443
AWGN	Condition-wise Sinkhorn	0.1154	0.4277
Rayleigh	Kernel target	0.3757	1.2869
Rayleigh	Kernel joint	0.1994	0.8198
Rayleigh	Joint Sinkhorn	0.1723	0.6929
Rayleigh	Condition-wise Sinkhorn	0.1097	0.3820
SSPA	Kernel target	0.2118	0.7631
SSPA	Kernel joint	0.2135	0.7685
SSPA	Joint Sinkhorn	0.2174	0.7876
SSPA	Condition-wise Sinkhorn	0.0566	0.2376
TDL	Kernel target	0.3155	1.1632
TDL	Kernel joint	0.2934	1.0939
TDL	Joint Sinkhorn	0.3221	1.1871
TDL	Condition-wise Sinkhorn	0.1054	0.4444

The SSPA condition-wise Sinkhorn row uses the same 30-seed update-budget-controlled run as Table V.

condition-wise Sinkhorn is the most reliable drifting variant and fixes the SSPA failure mode once the update budget is matched to the sharp same-condition field. The per-channel SER curves in Figs. 4–7 add WGAN and diffusion references to this downstream comparison. At the nominal training point, condition-wise Sinkhorn nearly reaches the analytic AWGN floor, while diffusion is the strongest learned baseline on Rayleigh, SSPA, and TDL. On SSPA, the $M_{\text{msg}} = 64$ update-budget screen gives condition-wise Sinkhorn SER 8.13×10^{-5} at $E_b/N_0 = 8$ dB, compared with 7.50×10^{-5} for WGAN, 3.13×10^{-5} for diffusion, and 1.57×10^{-5} for analytic-channel training.

For SSPA, we follow the diffusion-channel setup of [1] more closely and use message alphabet $M_{\text{msg}} = 64$, $n = 8$,

rate 6/8, and training at $E_b/N_0 = 8$ dB. The 30-seed SSPA screen in Table VIII shows that the corrected condition-wise Sinkhorn implementation is reliable once the W-Flow update budget is matched to the observed convergence of the sharp same-condition field. Figs. 4–7 report SER over an E_b/N_0 grid for AWGN, Rayleigh, SSPA, and TDL, which also shows where diffusion retains an advantage over one-shot drifting in downstream coding performance.

D. Long-block TurboAE check

The symbolic autoencoder results above use short code-words, so we additionally run a longer-block neural-coding check with a CNN TurboAE-style encoder–decoder at block length 64, rate 1/2, and AWGN training point $E_b/N_0 = 4$ dB. The TurboAE models are trained either through the analytic AWGN channel or through a condition-wise Sinkhorn AWGN surrogate and are then evaluated on the analytic AWGN channel. This experiment complements the short-block multi-channel SER/BER study. It checks whether the learned one-shot channel implant remains usable in a longer differentiable coding loop.

Fig. 8 shows the resulting BER and BLER curves over 30 seeds. At the 4 dB training point, analytic-channel training gives BER 1.96×10^{-3} with 95% CI 5.21×10^{-4} and BLER 8.18×10^{-2} with 95% CI 2.30×10^{-2} . Training through the condition-wise Sinkhorn surrogate gives BER 3.30×10^{-3} with 95% CI 4.30×10^{-4} and BLER 1.39×10^{-1} with 95% CI 1.91×10^{-2} . Thus the surrogate has a measurable long-block gap, but it does not cause a collapse of TurboAE training.

E. Equal wall-clock channel-implant training

The previous downstream tables compare final trained channel implants. Table IX instead fixes the symbolic autoencoder training time to 1800 s per implant and records how many optimizer updates fit inside that budget. This directly measures

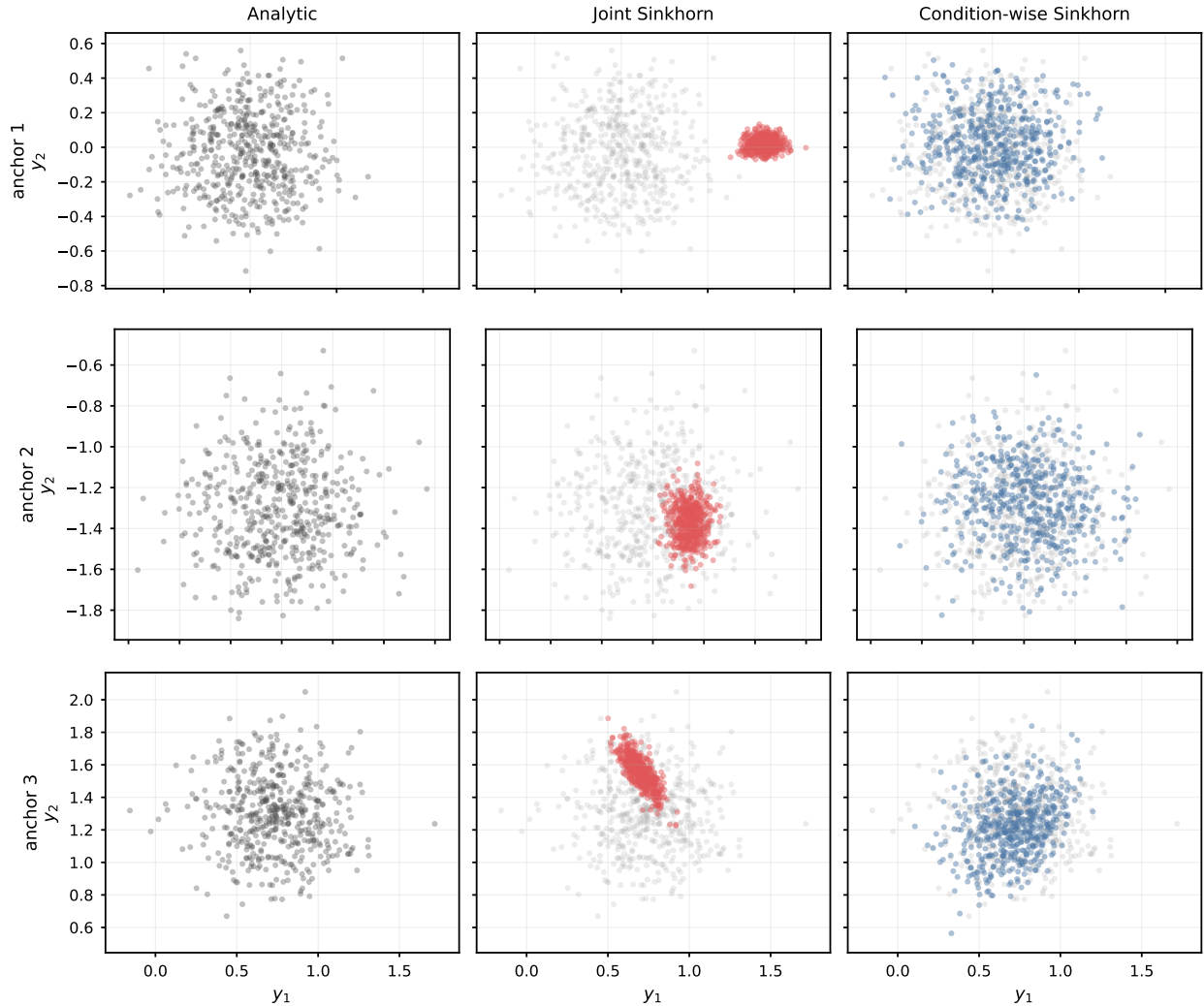


Fig. 3. **Fixed-condition SSPA output fibers.** Each row fixes one transmitted-symbol anchor x_i and plots repeated channel outputs in the first I/Q plane. The analytic channel is shown in gray. The learned columns overlay generated samples from joint Sinkhorn and condition-wise Sinkhorn on the same analytic cloud. Joint transport can match aspects of the global output distribution while displacing the fixed- x fiber. Condition-wise Sinkhorn better preserves the conditional output law.

the inner-loop setting that motivates one-shot channel surrogates. DDIM-100 provides a strong learned reference in final-performance comparisons, but its iterative sampler reduces the number of autoencoder updates by roughly one to two orders of magnitude relative to the one-shot implants. Under this equal wall-clock budget, condition-wise Sinkhorn is the best learned implant on AWGN and SSPA, DDIM-100 is best on Rayleigh, and WGAN is best on compact TDL. The result supports a limited claim: condition-wise Sinkhorn does not dominate diffusion or WGAN in every channel, but it reaches a competitive downstream operating point while retaining one-shot channel calls.

V. DISCUSSION

A. Accuracy-latency operating point

Condition-wise Sinkhorn drifting targets a low-latency conditional-surrogate operating point. Diffusion remains the strongest learned reference on several hard downstream SER

curves and pays for that fidelity through iterative sampling. The one-shot drifting path uses a single generator evaluation and preserves the fixed-input structure of the channel law. This makes it relevant when the learned channel is embedded in an inner training loop, where surrogate calls can dominate the computational budget. The main methodological lesson is that channel surrogates should be judged by conditional-law quality and downstream communication metrics in addition to global sample-cloud agreement.

B. Metric-space dependence

Direct and residual drifting behave differently across channel classes and evaluation spaces. Residual-space SWD is useful for checking a residual-mode generator and remains a secondary diagnostic for a channel simulator whose output is ultimately y . The W-Flow run reinforces this metric-dependence message. Kernel target, joint kernel, joint Sinkhorn, and condition-wise Sinkhorn can swap order depending on whether

TABLE VIII

DOWNSTREAM SYMBOLIC CODING METRICS FOR W-FLOW CHANNEL SURROGATES. AUTOENCODERS ARE TRAINED THROUGH EACH CHANNEL SURROGATE AND EVALUATED ON THE ANALYTIC CHANNEL. VALUES ARE MEAN $\pm$ STANDARD ERROR OVER SEEDS. LOWER IS BETTER. BOLD MARKS THE BEST LEARNED SURROGATE IN EACH ROW, WHILE THE ANALYTIC CHANNEL IS THE REFERENCE FLOOR.

Channel	Metric	Analytic	Kernel target	Kernel joint	Joint Sinkhorn	Condition-wise Sinkhorn
AWGN	BER	$0.00293 \pm 4.0 \times 10^{-5}$	$0.0187 \pm 5.7 \times 10^{-4}$	$0.0124 \pm 3.8 \times 10^{-4}$	$0.00388 \pm 5.5 \times 10^{-5}$	$0.00302 \pm 4.2 \times 10^{-5}$
AWGN	SER	$0.00551 \pm 7.3 \times 10^{-5}$	0.0351 ± 0.001	$0.0233 \pm 7.1 \times 10^{-4}$	$0.00729 \pm 9.7 \times 10^{-5}$	$0.00568 \pm 7.5 \times 10^{-5}$
Rayleigh	BER	$0.00273 \pm 4.3 \times 10^{-5}$	$0.0359 \pm 7.1 \times 10^{-4}$	0.111 ± 0.0017	$0.0177 \pm 3.3 \times 10^{-4}$	$0.00471 \pm 8.2 \times 10^{-5}$
Rayleigh	SER	$0.00514 \pm 8.0 \times 10^{-5}$	0.0685 ± 0.0012	0.208 ± 0.0032	$0.0331 \pm 5.6 \times 10^{-4}$	$0.00881 \pm 1.4 \times 10^{-4}$
SSPA	BER	$6.72 \times 10^{-6} \pm 1.9 \times 10^{-6}$	0.0661 ± 0.0046	–	–	$4.90 \times 10^{-5} \pm 7.5 \times 10^{-6}$
SSPA	SER	$1.30 \times 10^{-5} \pm 3.5 \times 10^{-6}$	0.134 ± 0.0092	–	–	$9.33 \times 10^{-5} \pm 1.4 \times 10^{-5}$
TDL	BER	$0.00158 \pm 2.1 \times 10^{-5}$	$0.0102 \pm 2.6 \times 10^{-4}$	$0.0098 \pm 2.5 \times 10^{-4}$	$0.0107 \pm 3.7 \times 10^{-4}$	$0.00293 \pm 4.4 \times 10^{-5}$
TDL	SER	$0.00297 \pm 3.6 \times 10^{-5}$	$0.0194 \pm 4.9 \times 10^{-4}$	$0.0185 \pm 4.6 \times 10^{-4}$	$0.02 \pm 6.3 \times 10^{-4}$	$0.00546 \pm 7.6 \times 10^{-5}$

SSPA uses the 30-seed $M_{\text{msg}} = 64$ update-budget screen. Dashes mark variants not rerun under that SSPA coding setup.

TABLE IX

EQUAL WALL-CLOCK SYMBOLIC AUTOENCODER TRAINING. EACH ROW TRAINS THE SAME SYMBOLIC AUTOENCODER FOR 1800 s THROUGH THE LISTED CHANNEL IMPLANT AND EVALUATES THE RESULTING ENCODER/DECODER ON THE ANALYTIC CHANNEL. RESULTS ARE MEAN $\pm$ STANDARD ERROR OVER 30 SEEDS.

Channel	Training implant	Updates [k]	SER	BER
AWGN	Analytic	705.7	$3.54 \times 10^{-3} \pm 3.94 \times 10^{-5}$	$1.89 \times 10^{-3} \pm 2.31 \times 10^{-5}$
	Condition-wise Sinkhorn	626.8	$3.62 \times 10^{-3} \pm 3.65 \times 10^{-5}$	$1.93 \times 10^{-3} \pm 2.31 \times 10^{-5}$
	WGAN	641.8	$3.70 \times 10^{-3} \pm 4.06 \times 10^{-5}$	$1.98 \times 10^{-3} \pm 2.87 \times 10^{-5}$
	DDIM-100	21.5	$5.32 \times 10^{-3} \pm 1.36 \times 10^{-4}$	$2.83 \times 10^{-3} \pm 6.82 \times 10^{-5}$
Rayleigh	Analytic	656.3	$2.25 \times 10^{-3} \pm 4.57 \times 10^{-5}$	$1.22 \times 10^{-3} \pm 2.94 \times 10^{-5}$
	Condition-wise Sinkhorn	602.1	$5.40 \times 10^{-3} \pm 1.11 \times 10^{-4}$	$2.92 \times 10^{-3} \pm 7.00 \times 10^{-5}$
	WGAN	614.0	$4.83 \times 10^{-3} \pm 2.87 \times 10^{-4}$	$2.55 \times 10^{-3} \pm 1.52 \times 10^{-4}$
	DDIM-100	21.3	$4.64 \times 10^{-3} \pm 1.08 \times 10^{-4}$	$2.47 \times 10^{-3} \pm 6.10 \times 10^{-5}$
SSPA	Analytic	635.6	$1.33 \times 10^{-6} \pm 6.31 \times 10^{-7}$	$7.78 \times 10^{-7} \pm 3.81 \times 10^{-7}$
	Condition-wise Sinkhorn	617.1	$8.80 \times 10^{-5} \pm 1.87 \times 10^{-5}$	$4.36 \times 10^{-5} \pm 8.30 \times 10^{-6}$
	WGAN	627.3	$9.83 \times 10^{-5} \pm 3.15 \times 10^{-5}$	$4.50 \times 10^{-5} \pm 1.27 \times 10^{-5}$
	DDIM-100	21.4	$9.00 \times 10^{-5} \pm 4.64 \times 10^{-5}$	$4.03 \times 10^{-5} \pm 1.72 \times 10^{-5}$
TDL	Analytic	449.0	$1.30 \times 10^{-3} \pm 1.98 \times 10^{-5}$	$6.98 \times 10^{-4} \pm 1.19 \times 10^{-5}$
	Condition-wise Sinkhorn	576.8	$3.44 \times 10^{-3} \pm 6.05 \times 10^{-5}$	$1.82 \times 10^{-3} \pm 3.43 \times 10^{-5}$
	WGAN	587.6	$3.21 \times 10^{-3} \pm 4.36 \times 10^{-4}$	$1.70 \times 10^{-3} \pm 2.19 \times 10^{-4}$
	DDIM-100	21.3	$4.64 \times 10^{-3} \pm 1.23 \times 10^{-4}$	$2.47 \times 10^{-3} \pm 6.36 \times 10^{-5}$

Bold marks the best learned implant per channel; the analytic row is the non-learned reference. The DDIM row uses 100 denoising steps per channel call, which yields many fewer optimizer updates within the same autoencoder-training time budget.

the metric is global SWD, anchor-conditioned SWD, or downstream BER/SER. We therefore report direct-output SWD, anchor-conditioned moment metrics, and downstream coding metrics side by side. Hyperparameter selection from a single global SWD score would miss these differences.

C. Conditional Sinkhorn scope

The condition-wise estimator is directly applicable to analytic or simulator channels because repeated outputs can be drawn at the same transmitted symbol. Measured channels are harder. With only one observation per x , the exact conditional law $p(\cdot | x)$ cannot be observed at a point. For that regime, the appropriate practical variant is local conditional Sinkhorn,

where nearby conditions are grouped or kernel-weighted and the Sinkhorn problem is solved only inside a local condition block. Practical channel simulators may also need transport features that encode phase, amplitude, residual structure, or decoder-relevant projections.

VI. CONCLUSION

This paper makes one-shot drifting applicable to learned channel simulation by replacing unconditional transport with condition-wise Sinkhorn transport over fixed-input output fibers. The resulting generator preserves the transmitted symbol x , transports only the conditional output law, and keeps the one-shot inference path that motivates drifting for inner-loop

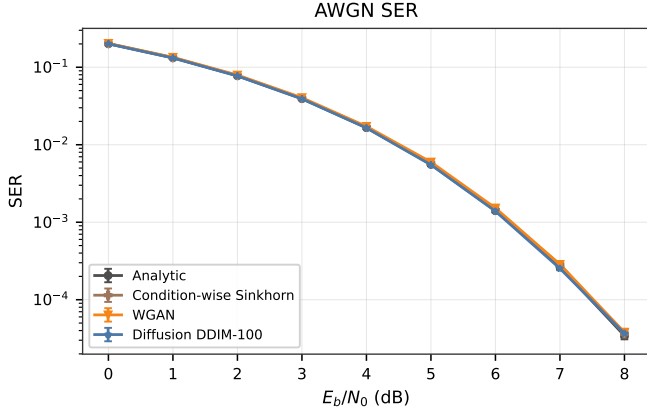


Fig. 4. **AWGN SER curve.** Symbolic autoencoders are trained through analytic, condition-wise Sinkhorn, WGAN, or DDIM-100 channel implants and evaluated on the analytic channel. Points show seed means with standard-error bars.

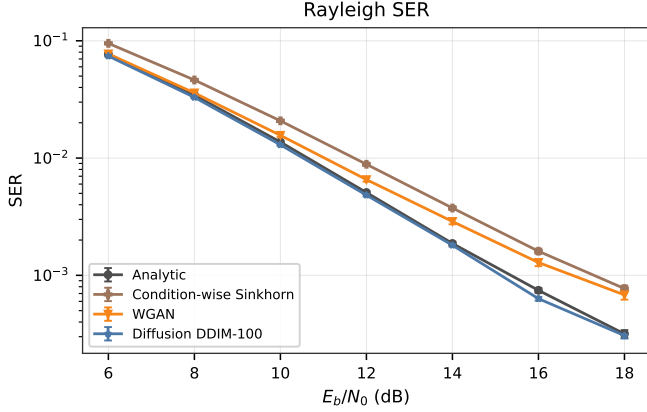


Fig. 5. **Rayleigh SER curve.** Symbolic autoencoders are trained through analytic, condition-wise Sinkhorn, WGAN, or DDIM-100 channel implants and evaluated on the analytic channel. Points show seed means with standard-error bars.

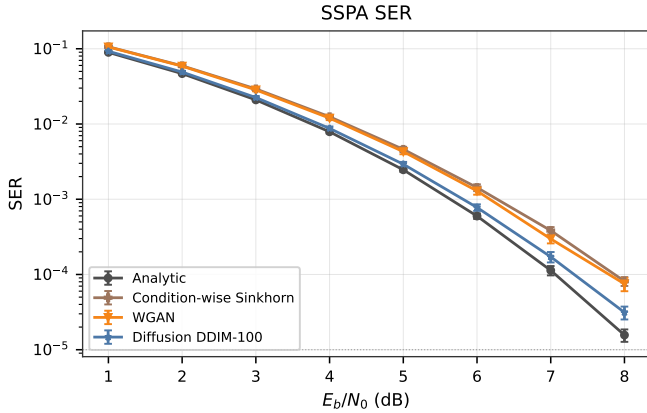


Fig. 6. **SSPA SER curve.** Symbolic autoencoders are trained through analytic, condition-wise Sinkhorn, WGAN, or DDIM-100 channel implants and evaluated on the analytic channel. Points show seed means with standard-error bars.

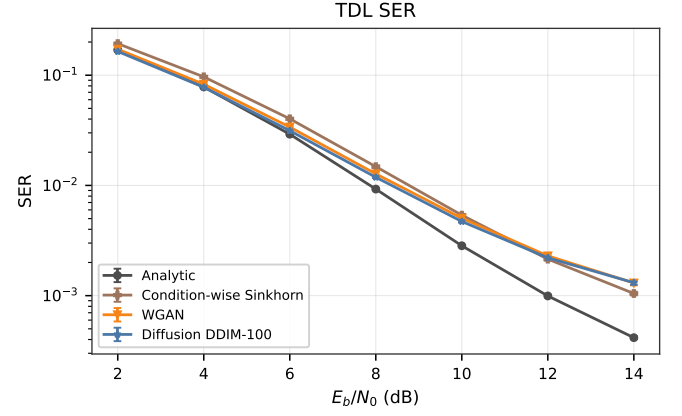


Fig. 7. **TDL SER curve.** Symbolic autoencoders are trained through analytic, condition-wise Sinkhorn, WGAN, or DDIM-100 channel implants and evaluated on the analytic channel. Points show seed means with standard-error bars.

of $p(\cdot | x)$ and $q_\theta(\cdot | x)$ for μ -almost every x as the correct equilibrium geometry. The implemented method is a finite-sample barycentric approximation followed by projected detached-target neural training.

Experiments on AWGN, Rayleigh, SSPA, and compact TDL channels show a clear accuracy-latency tradeoff. Within the matched drifting-family ablation, condition-wise Sinkhorn is the best-performing one-shot surrogate under the downstream symbolic-coding checks. Diffusion remains the strongest learned reference on the hardest SER curves. This positions the contribution as a low-latency, condition-preserving operating point on the accuracy-latency frontier. The SSPA study also shows that optimizer budget is part of the drift-field design. The corrected same-condition field works well after convergence, while the full diffusion-style preset can over-optimize this sharper one-step field. Across the full set of results, global SWD, anchor-conditioned metrics, and downstream coding metrics can disagree. This makes metric choice a central part of learned channel simulation and motivates future work on channel-specific transport features and local conditional Sinkhorn approximations for measured channels.

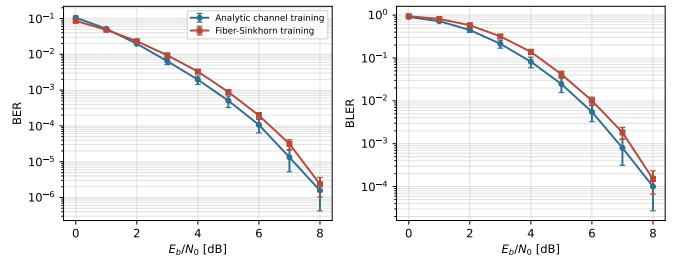


Fig. 8. **Long-block TurboAE AWGN check.** CNN TurboAE models are trained at block length 64, rate 1/2, and $E_b/N_0 = 4$ dB, either through the analytic AWGN channel or through the condition-wise Sinkhorn AWGN surrogate. Curves are evaluated on the analytic AWGN channel over 30 seeds. The surrogate preserves the qualitative waterfall trend but remains above the analytic-channel training curve.

channel use. The population formulation identifies equality

REFERENCES

- [1] M. Kim, R. Fritschek, and R. F. Schaefer, “Diffusion models for accurate channel distribution generation,” *arXiv preprint arXiv:2309.10505*, 2023.
- [2] —, “Robust generation of channel distributions with diffusion models,” in *IEEE International Conference on Communications (ICC)*, 2024.
- [3] T. Lee, J. Park, H. Kim, and J. G. Andrews, “Generating high dimensional user-specific wireless channels using diffusion models,” *arXiv preprint arXiv:2409.03924*, 2024.
- [4] X. Gong, X. Liu, A. A. Lu, X. Gao, X. G. Xia, C.-X. Wang, and X. You, “Digital twin of channel: Diffusion model for sensing-assisted statistical channel state information generation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 5, pp. 3805–3821, 2025.
- [5] X. Zhou, L. Liang, J. Zhang, P. Jiang, Y. Li, and S. Jin, “Generative diffusion models for high dimensional channel estimation,” *IEEE Transactions on Wireless Communications*, vol. 24, no. 7, pp. 5840–5854, 2025.
- [6] B. Fesl, M. Baur, F. Strasser, M. Joham, and W. Utschick, “Diffusion-based generative prior for low-complexity MIMO channel estimation,” *IEEE Wireless Communications Letters*, vol. 13, no. 12, pp. 3493–3497, 2024.
- [7] Z. Chen, H. Shin, and A. Nallanathan, “Generative diffusion model-based variational inference for MIMO channel estimation,” *IEEE Transactions on Communications*, vol. 73, no. 10, pp. 9254–9269, 2025.
- [8] N. Zilberstein, A. Swami, and S. Segarra, “Joint channel estimation and data detection in massive MIMO systems based on diffusion models,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
- [9] H. Fu, W. Si, and R. Liu, “Conditional denoising diffusion-based channel estimation for fast time-varying MIMO-OFDM systems,” *Digital Signal Processing*, vol. 164, p. 105283, 2025.
- [10] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.
- [11] Y. Song, P. Dhariwal, M. Chen, and I. Sutskever, “Consistency models,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 32 211–32 252. [Online]. Available: <https://proceedings.mlr.press/v202/song23a.html>
- [12] T. Yin, M. Gharbi, R. Zhang, E. Shechtman, F. Durand, W. T. Freeman, and T. Park, “One-step diffusion with distribution matching distillation,” in *CVPR*, 2024.
- [13] M. Deng, H. Li, T. Li, Y. Du, and K. He, “Generative modeling via drifting,” *arXiv preprint arXiv:2602.04770*, 2026.
- [14] S. Sahraee-Ardakan, M. Delbracio, and P. Milanfar, “The geometry of noise: Why diffusion models don’t need noise conditioning,” *arXiv preprint arXiv:2602.18428*, 2026.
- [15] J. Han, P. Li, Q. Guo, R. Xu, S. Ermon, and E. J. Candès, “One-step generative modeling via wasserstein gradient flows,” *arXiv preprint arXiv:2605.11755*, 2026.
- [16] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” in *Advances in Neural Information Processing Systems*, vol. 26, 2013, pp. 2292–2300.
- [17] J. Feydy, T. Séjourné, F.-X. Vialard, S.-i. Amari, A. Trounev, and G. Peyré, “Interpolating between optimal transport and MMD using sinkhorn divergences,” in *Proceedings of the Twenty-Second International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 89. PMLR, 2019, pp. 2681–2690. [Online]. Available: <https://proceedings.mlr.press/v89/feydy19a.html>
- [18] 3GPP, “Study on Channel Model for Frequencies from 0.5 to 100 GHz,” 3GPP, Technical Report TR 38.901, 2022, version 17.1.0.